Final Project – Deep Neural Networks (CS/DS 541, Murai, Fall 2026)

1 Introduction & Scope

For the remainder of the course, you will define and conduct a small-scale research project consisting of some machine learning problem that you will tackle using neural networks as the computational framework. The topic is mostly up to you, but you will need to submit a brief proposal (see below) describing what it is you are trying to accomplish. In addition to defining the problem, implementing neural network(s) to try to solve it, and conducting computational experiments, you will also need to write a short (4-5 pages, depending on your group size) describing your work.

All of the following kinds of projects are acceptable:

- **New domain**: Tackling a novel (i.e., never before tried) machine learning problem using neural networks.
- **Old domain, new approach**: Devising a novel neural network architecture to try to outperform an existing baseline on a machine learning problem.
- **Fancy network implementation**: Implementing and training a non-trivial neural network architecture yourself **from scratch** (using any standard neural networks software package such as TensorFlow, PyTorch, JAX, etc.).

You may **not** do a project on handwritten digit recognition using the MNIST dataset – it's time for something new.

2 Proposal

By 6:00pm on Thursday, October 22, 2026, please submit a concise and precise **3-4 paragraph** description of the problem you want to tackle. In addition, you must include answers to the following questions:

1. Who are the members of the project team (the team size is 4; in rare circumstances, 3-person teams will be considered), and how will the work of the project be distributed among the individual team members (it is OK if this changes later).
2. For each member, list existing knowledge potentially relevant to the project (beyond what was covered in class) and whether they plan to study and learn more about specific topics to complete the project successfully.[1]
3. **Make sure you discuss how and how often you will be communicating with each other during the collaboration**, e.g., in-person, Zoom, Slack, fax, etc. Every member must agree to the terms.
4. What **computational task** (e.g., semantic segmentation, audio generation, sentiment analysis for text, chemical formula classification, etc.) are you tackling, and what do you want to accomplish.
5. What are the **data** you have available for training. What do they consist of (inputs, target labels), how many do you have, how much diversity do they exhibit, etc.
6. How will you measure performance of your neural network. In particular, you should define a **reasonable baseline** that you are trying to beat. If the problem has never before been tackled and it's unclear whether a neural network could be useful at all, then the baseline might consist of just "guessing" the target labels (e.g., the mean value for regression, or a uniform distribution for classification). If there's

[1] Rationale: I want to account for how much the student has learned rather than just how much they leveraged current expertise).

already a lot of prior research, then outperforming some well-known machine learning approach (e.g., ImageNet) would be reasonable.

7. What **existing work** are you building on? Clearly, it would no suffice for a final project to take someone else's GitHub repo which exactly solves the problem you're solving, just run it, and then produce some results. **What you do for your final project must extend significantly beyond someone else's off-the-shelf work that you can just download.** Your proposal must make clear what the "delta" of your project is.

8. Does the project overlap with work that any of team members are doing OR have done as part of another course assignments? If so, they must attach the proposal submitted to the corresponding course along with the instructor contact information. **Full disclosure is required**. There must be a clear identification of which elements of the project should be considered for evaluation in each course.

For all points above, it is crucial that you **elaborate** and **be detailed**. Do not simply write (for example), "We will explore semantic segmentation of images." Instead, describe which specific techniques you will use, e.g., "We will consider the modification XYZ (explain) to the standard W-Net approach to semantic segmentation. The motivation is that XYZ can better achieve ABC because of MNO."

I will provide feedback on your project proposal. You **must** obtain approval from me before proceeding with the project.

3 Paper

As described in class, every team is required to produce a project report. See `CS541FinalProject.zip` in the Files section of Canvas for more details. It is fine (and recommended) to use the LaTeX template; however, if you prefer another text-editing environment (e.g., Google Docs, Microsoft Word), that is fine too – **just make sure that your paper looks very similar (in formatting and structure) to the template I provided**.

Each group should produce a **4-page report**. These page limits do not include references or appendices, which can extend up to infinite pages. There is no guarantee that the appendix will be thoroughly read, so make sure that your key experiments and results are in the first four pages, and point to specific sections of the Appendix for things that we should check out.

4 Presentation

Your team will give a short presentation of its project during class on **Monday, November 30** and **Thursday, December 3, 2026**. The project presentation is worth **5%** of your final course grade, and is evaluated on how well it motivated the topic, how clearly it explained your approach to tackling the problem, and how clearly it presented your results.

5 Software

In contrast to the previous homework assignments, you are free to use whatever neural network library you wish, e.g., TensorFlow, PyTorch, Keras, JAX, etc.

The use of GitHub repositories from the get-go are highly encouraged. This allows the course staff to assess individual contributions.

6 Deliverables

This semester we will use scaffolding to ensure that teams will make continuous progress on their projects. Hence, we will break down the project into smaller deliverables culminating in the presentation and final

report:

1. Proposal of a significant deep learning project (Oct 22)
2. Reply to comments on proposal (Oct 29)
3. An intermediate progress report (Nov 12)
4. Reply to comments on report (Nov 19)[2]
5. Project presentation (Nov 30 & Dec 3)
6. Paper write-up (Dec 10)

Your final project submission will consist of (1) all your code that you used to implement your project and conduct experiments; (2) your final report.presentation.

7 Teams

You are required to form a team of 3-4 people.

8 Grading

Each of the deliverables will receive separate grades. The project is worth **35%** of your final course grade, broken down as follows: Proposal (**3%**), Reply to proposal comments (**2%**), Intermediate progress report (**5%**), Project presentation (**5%**), and Project write-up & team contributions (**20%**).

The maximum raw score on the **Project write-up & team contributions** deliverable is **85 points**, divided into: (i) **70 points** for the report content, detailed below (Methodology, Performance, Thoroughness, Writing); (ii) **10 points** based on teammate evaluations; and (iii) **5 points** for individual growth. Criteria (ii) and (iii) will be evaluated based on surveys to be filled out by team members individually.

Detailed rubric for the report (70 points)

Each sub-criterion below is scored on its own scale and multiplied by the listed weight to get its (maximum) point contribution. When a category's sub-criteria add up to more than the category's point value, the category score is capped at that value – so your project does not need a perfect score on every sub-criterion to earn full credit in that category.

- **Methodology (20 points):** What procedures did you use to design and optimize both the parameters and hyperparameters of your network?
  - Design (scale 1–4, weight ×2, up to 8 pts): 1 = significant issues in the design; 2 = solid, with minor issues; 3 = solid, or sophisticated with minor issues; 4 = sophisticated.
  - Parameter optimization (scale 1–4, weight ×2, up to 8 pts): 1 = significant issues; 2 = reasonable, but loss curves not shown or discussed; 3 = did not fully converge, due to compute constraints; 4 = excellent.
  - Hyperparameter optimization (scale 1–3, weight ×2, up to 6 pts): 1 = not yet explored or discussed; 2 = one hyperparameter explored; 3 = two or more hyperparameters explored, or replicated despite cost.

  (Sub-criteria sum to 22 pts; category capped at 20.)

[2]This reply has no separate grade weight of its own, but is a required checkpoint: address the feedback you received on your progress report before you resume working on the project.

- Performance (**15 points**): How much more accurate, relative to the difficulty of the problem you defined, did your network get?
  - Performance (scale 1–4, weight ×3.75, up to 15 pts): 1 = significant issue; 2 = no comparison provided, or lower accuracy & speed, or a single evaluation metric; 3 = trade-off could be improved, or baseline could be stronger; 4 = higher accuracy, or a good accuracy/speed trade-off.
- Thoroughness (**20 points**): Obtaining one research result often opens up many more questions. Does your project offer clear "lessons learned", or does it simply report a few numbers with little insight? Were there obvious research questions that could have easily been explored but weren't?
  - Problem solving (scale 0–2, weight ×4, up to 8 pts; N/A if not applicable): 0 = faced challenges that were not resolved; 1 = made some effort to tackle challenges; 2 = went above and beyond.
  - Experiments (scale 1–3, weight ×4, up to 12 pts): 1 = basic set of experiments; 2 = adequate; 3 = comprehensive.
  - Analysis (scale 0–3, weight ×4, up to 12 pts): 0 = plots and discussion not yet included; 1 = plots included, with limited discussion; 2 = some discussion; 3 = deep insights.

  (Sub-criteria sum to 32 pts; category capped at 20.)
- Writing (**15 points**): Is the paper understandable? Are all important details clear? Is it well organized? Does the paper cite relevant prior research and explain the significance of the results relative to the prior state-of-the-art? (Note: for "fancy network implementation" projects, make sure that you relate the network you are implementing to prior architectures and motivate why it is computationally interesting, effective, or promising.)
  - Presentation of the writing (scale 1–4, weight ×2, up to 8 pts): 1 = needs improvement; 2 = fair; 3 = good; 4 = excellent.
  - Details/reproducibility (scale 1–4, weight ×2, up to 8 pts): 1 = difficult to reproduce from the details given; 2 = some important details missing; 3 = difficult to reproduce, but due to lack of space; 4 = great.
  - Related work (scale 1–3, weight ×2, up to 6 pts): 1 = related work section could be expanded; 2 = OK, but incomplete; 3 = excellent.

  (Sub-criteria sum to 22 pts; category capped at 15.)
- Team evaluations (**10 points**): Has your team attempted to balance the work equitably between its members? Can you provide a range of the average number of hours per week spent by the team members? If you are using a GitHub repo, can you quantify their contributions? Note: having a member that is too enthusiastic about the project does not hurt the other members, as long as they have also contributed to the work. IMPORTANT NOTE: The KEY to get good team evaluations is GREAT COMMUNICATION. As a team member, you need to strive to set your team's expectations, define due dates for your deliverables and try your best to make them, or otherwise communicate roadblocks in advance and get help. Self-evaluations or team member evaluations that are too disparate from the rest (either much more positive or much more negative) will be disregarded.
- Individual Growth (**5 points**): What have you learned in this project in terms of deep learning techniques?

To earn a good grade, please make sure that your paper clearly describes your methodology and empirical results (to evaluate performance), and highlights your thoroughness in examining your research problem. Note that, for a course-based project such as this one, **negative results are acceptable**. While desirable, it is **not** necessary to obtain excellent performance in order to earn a good grade. As long as your methodology is sound and you conducted a thorough (within the confines of a 6-week project) exploration of your target domain, you can still earn an A.